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(54) **HETEROGENEOUS FILAMENTS, METHODS OF PRODUCING THE SAME, SCAFFOLDS, METHODS OF PRODUCING THE SAME, DROPLETS, AND METHODS OF PRODUCING THE SAME**

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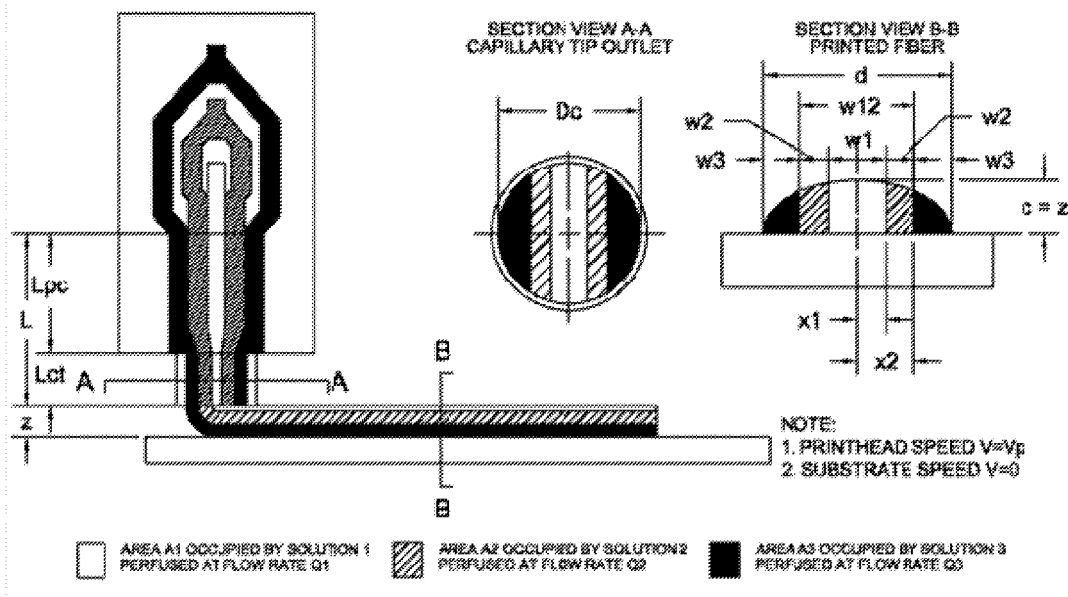
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(57)

ABSTRACT

One aspect of the invention provides a method of producing a heterogeneous filament. The method includes applying pressure to a plurality of filament components to move the filament components into a common outlet channel. Another aspect of the invention provides a method of fabricating a scaffold. The method includes producing a plurality of filaments according to the methods described herein. Another aspect of the invention provides a scaffold produced according to the methods described herein. Another aspect of the invention provides a heterogeneous filament including a plurality of adjacent filament components. Another aspect of the invention provides a method of producing a droplet. The method includes: positioning a printing tip in proximity to a cross-linking liquid; applying pressure to one or more liquids to produce a droplet extending beyond outside of the printing tip; and contacting the droplet with the cross-linking liquid, thereby producing a droplet.



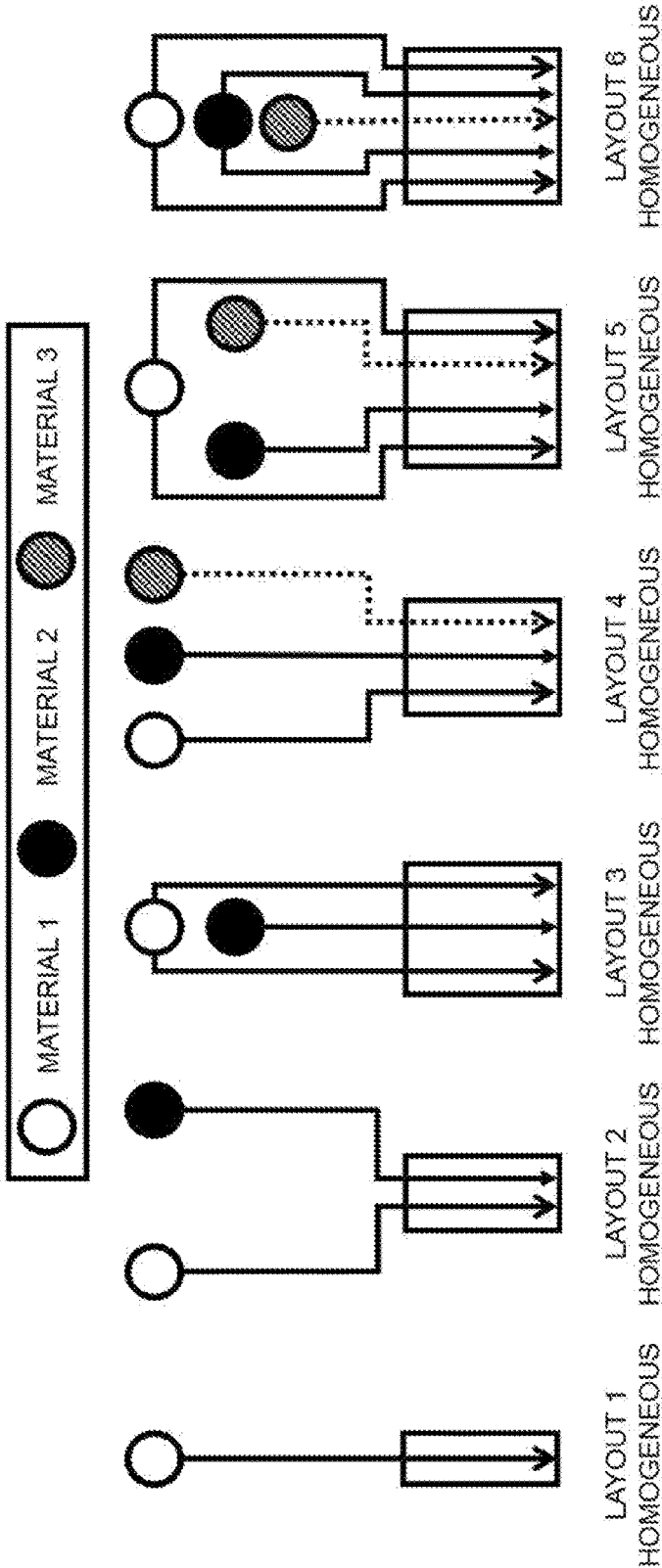


FIG. 1

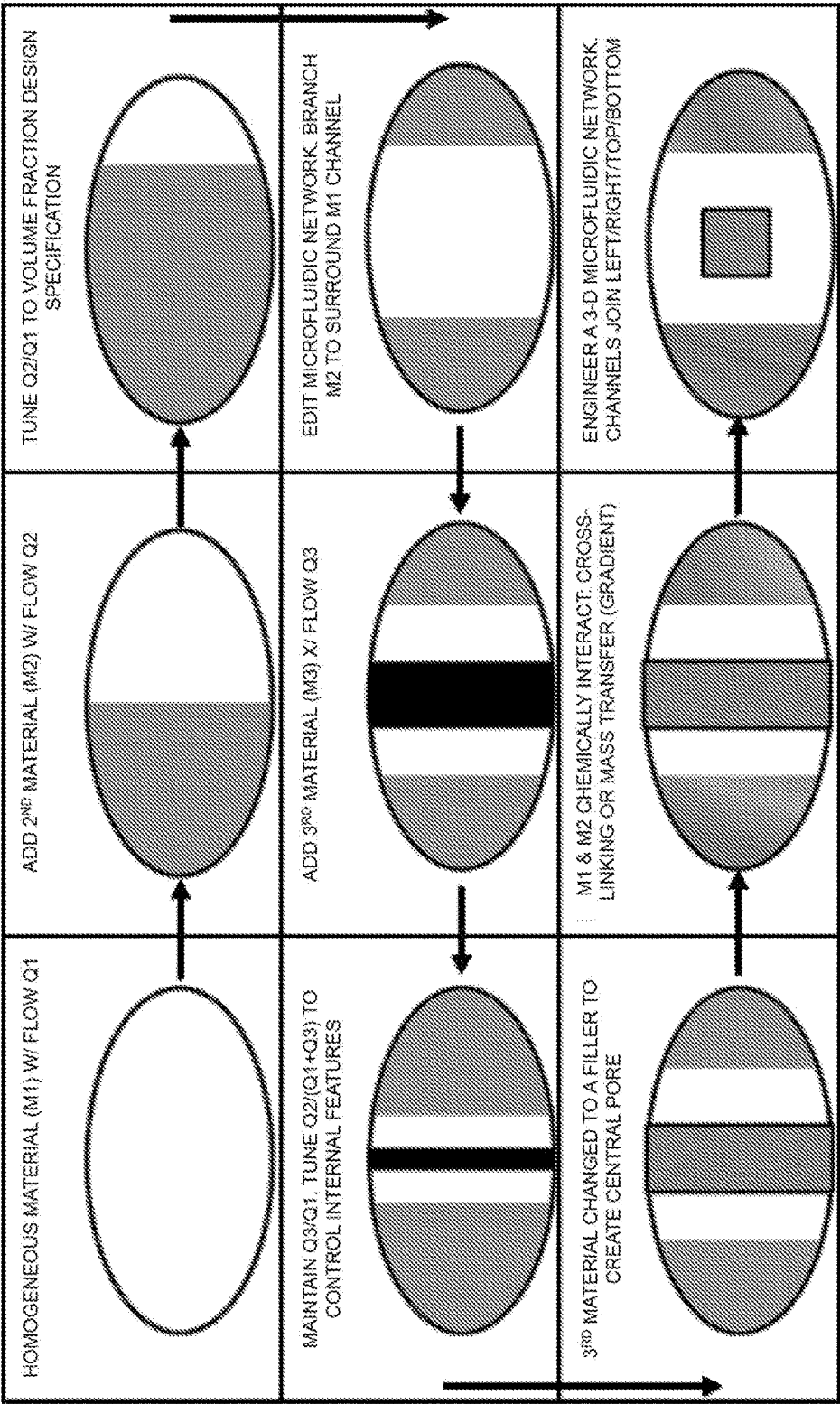


FIG. 2

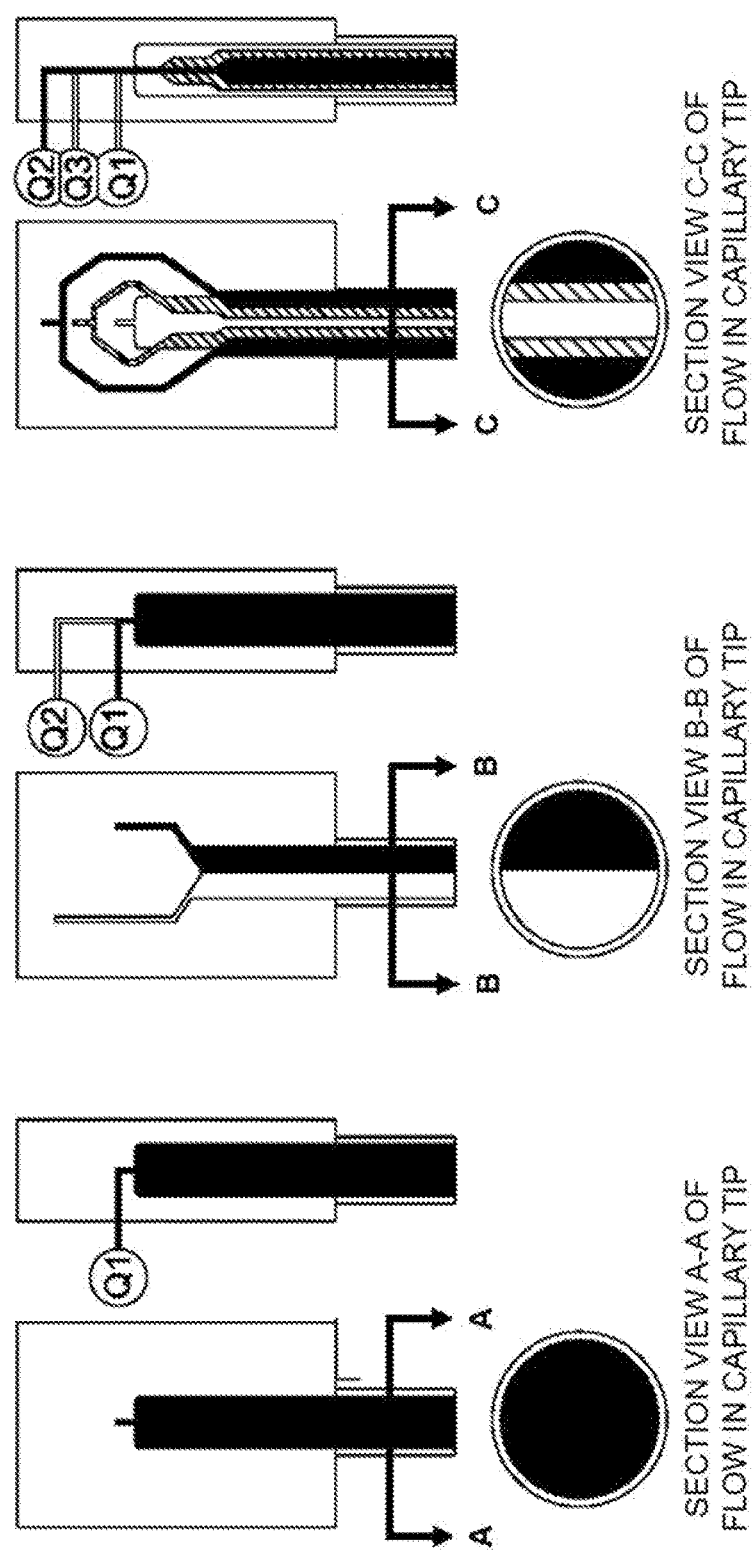


FIG. 3

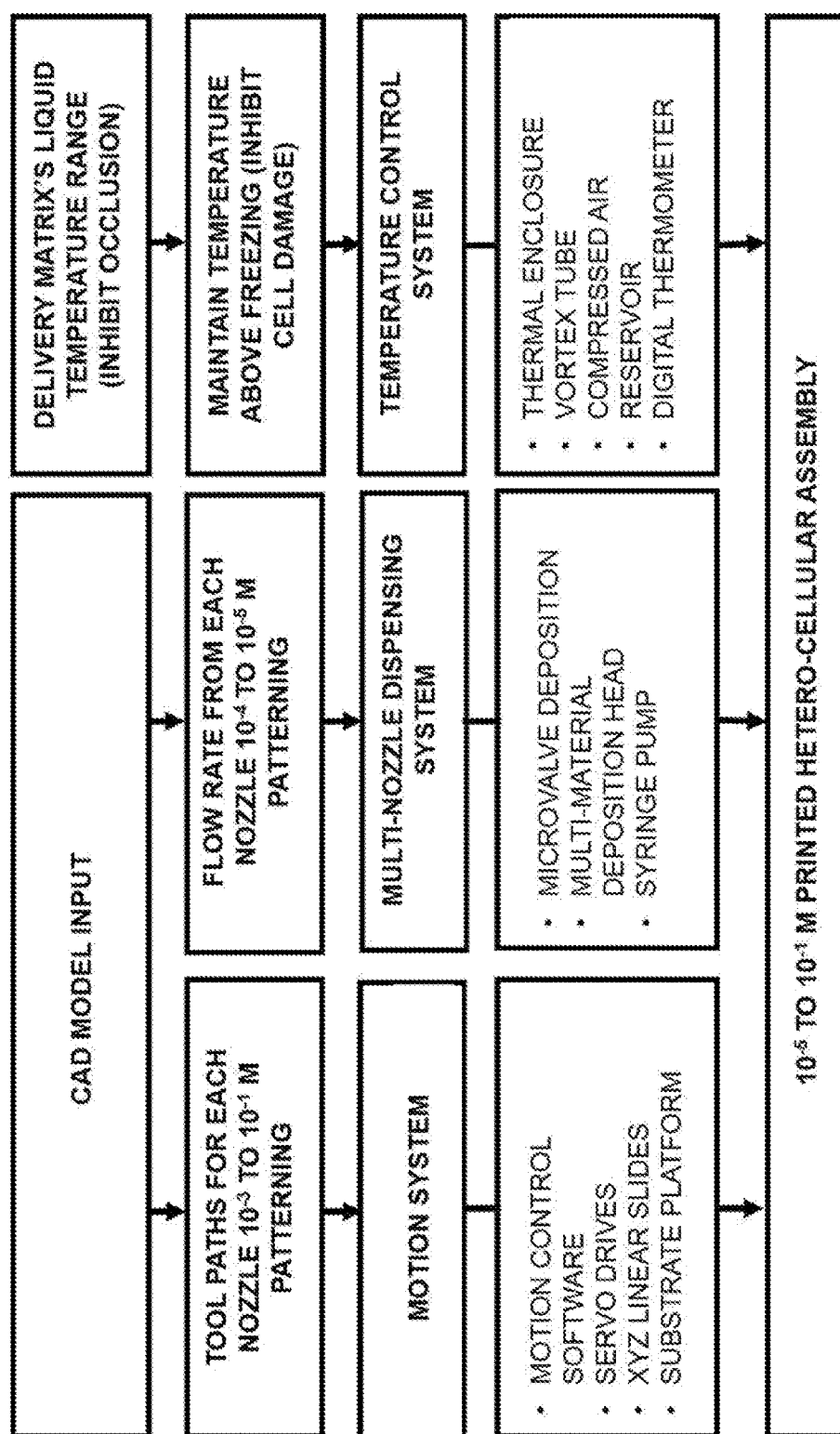


FIG. 4

1-DIMENSIONAL ARRAY OF SOLUTIONS IN OUTLET CHANNEL

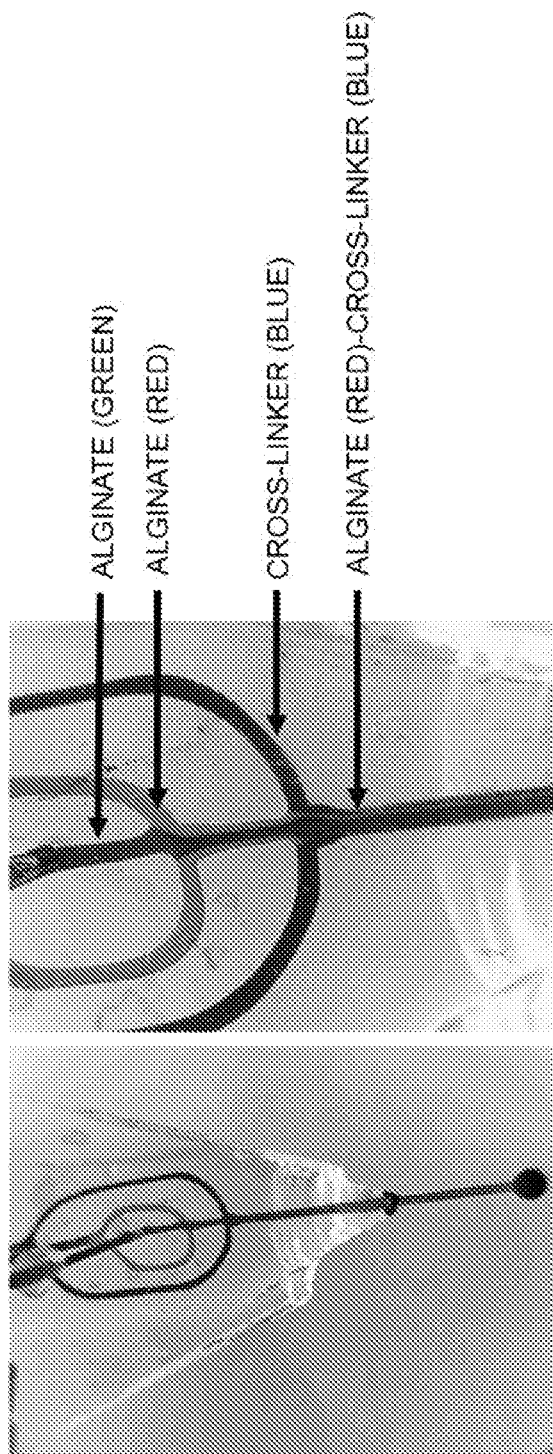


FIG. 5

- Step 1.
Remove PCL from PDMS
- Step 2.
Shape outlet to a point
- Step 3.
Insert $\phi 250\mu\text{m}$ capillary tip into outlet
- Step 4.
Insert capillary tip into inlets.
Connect inlet tips to silicone tubes

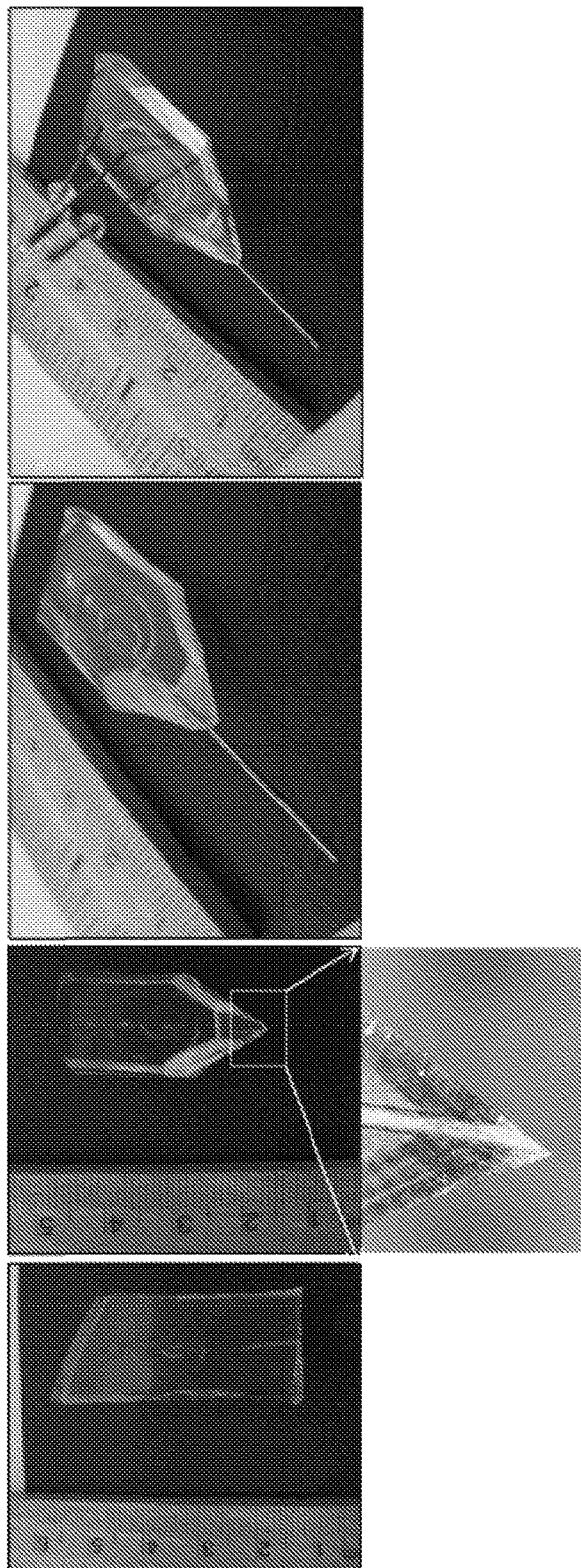


FIG. 6

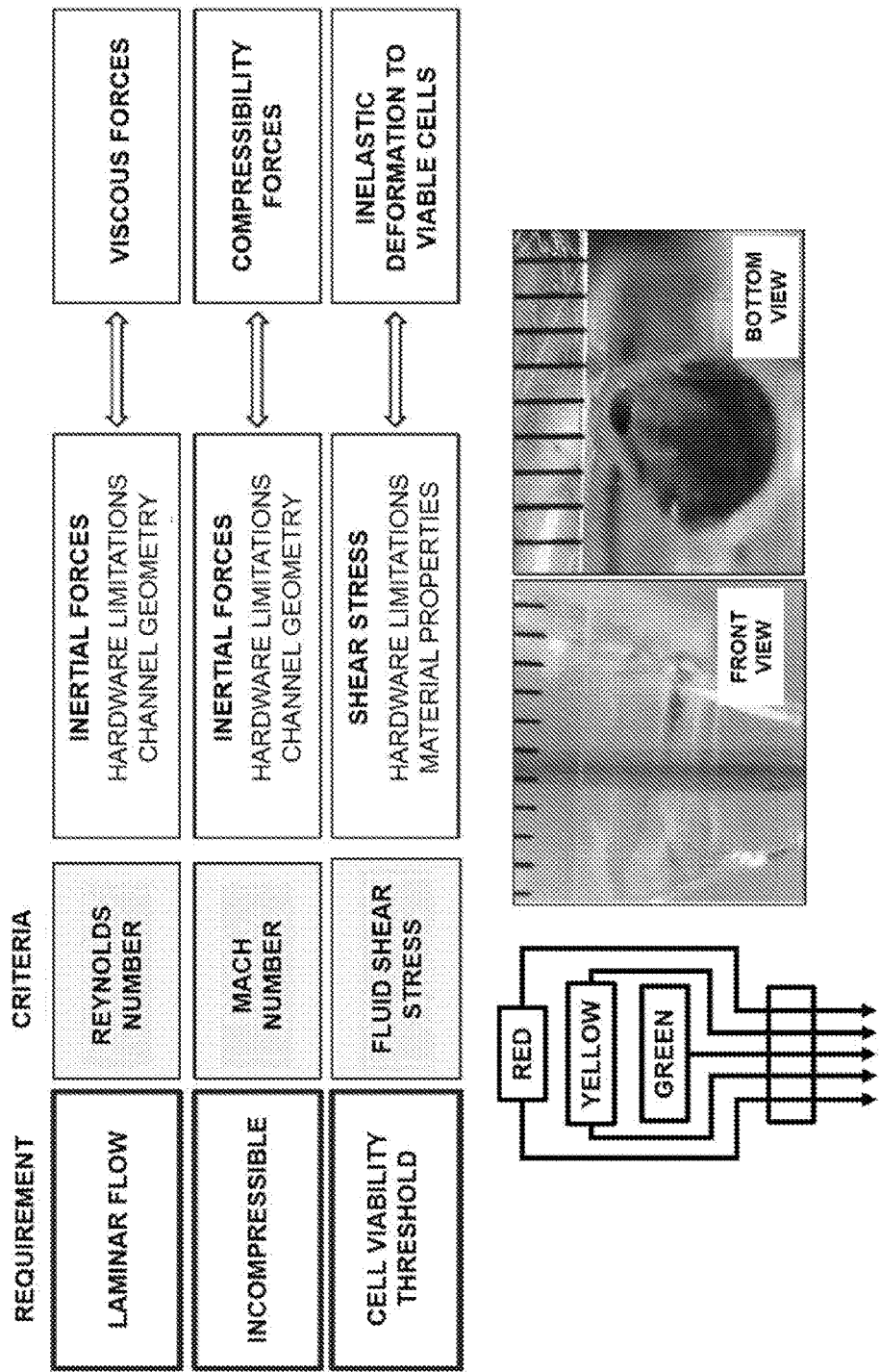


FIG. 7

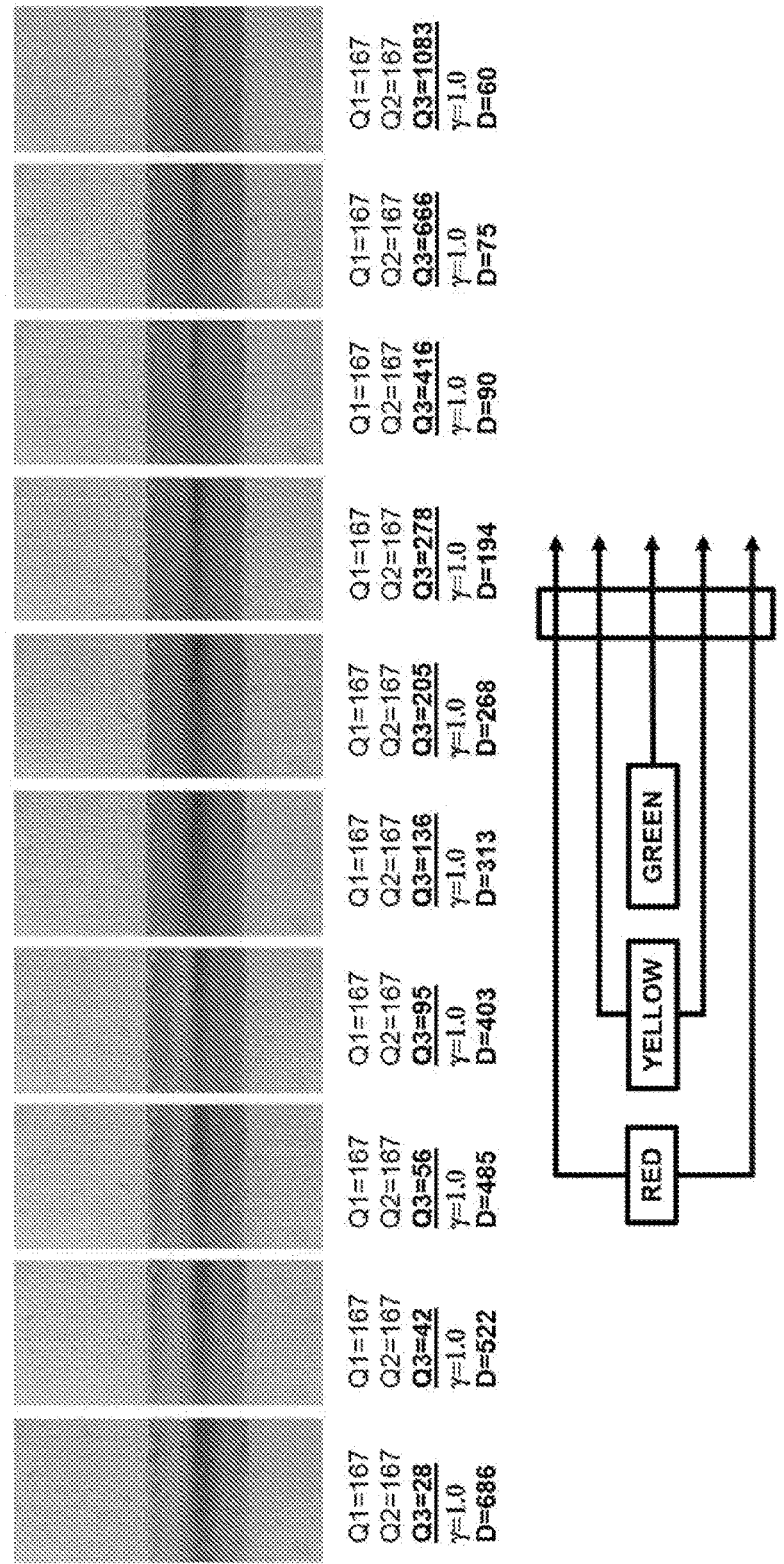


FIG. 8

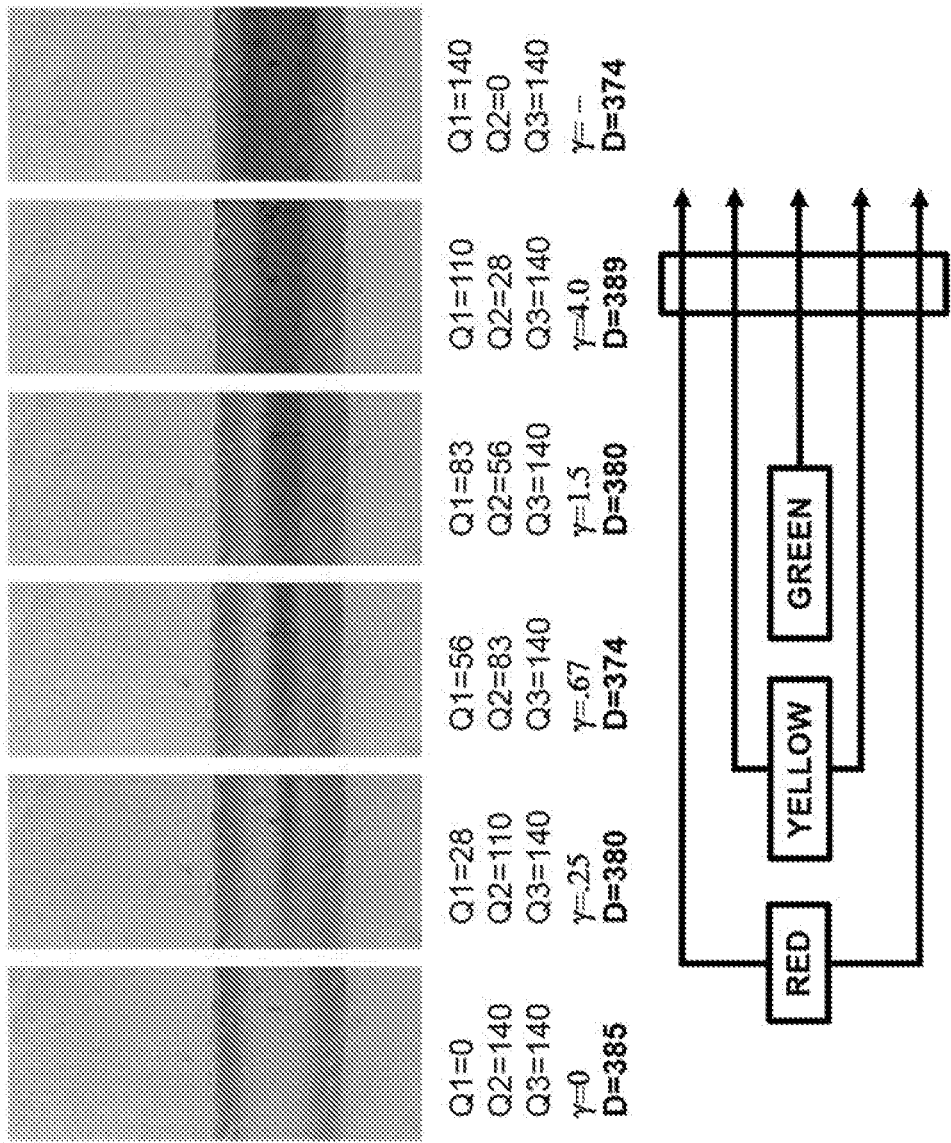
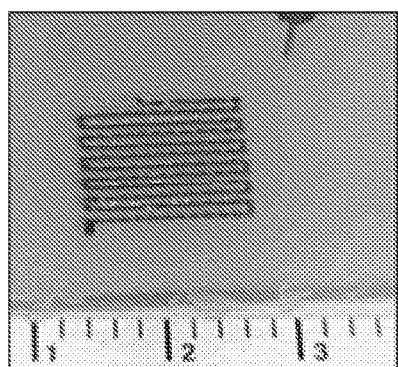
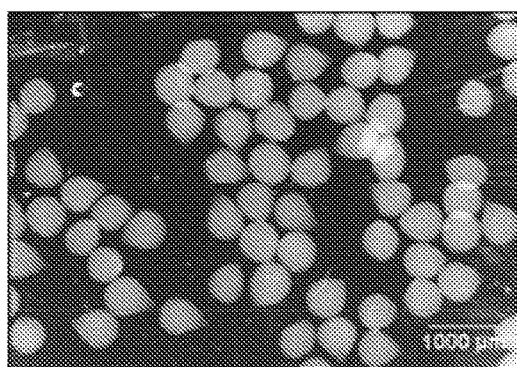


FIG. 9



*Extrusion along a tool path to
print scaffold.*



*Control the frequency of an
event to print droplets.*

FIG. 10

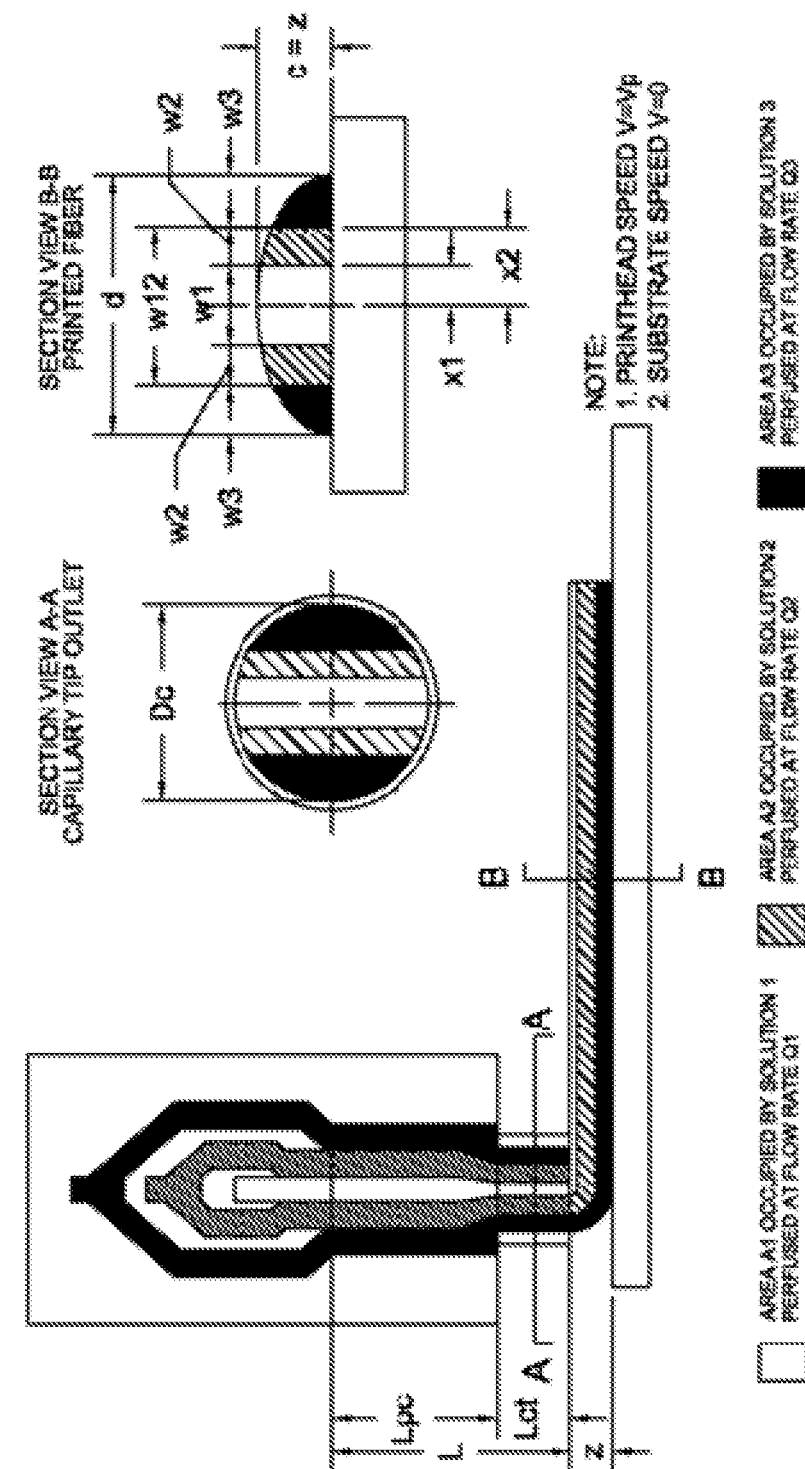


FIG. 11

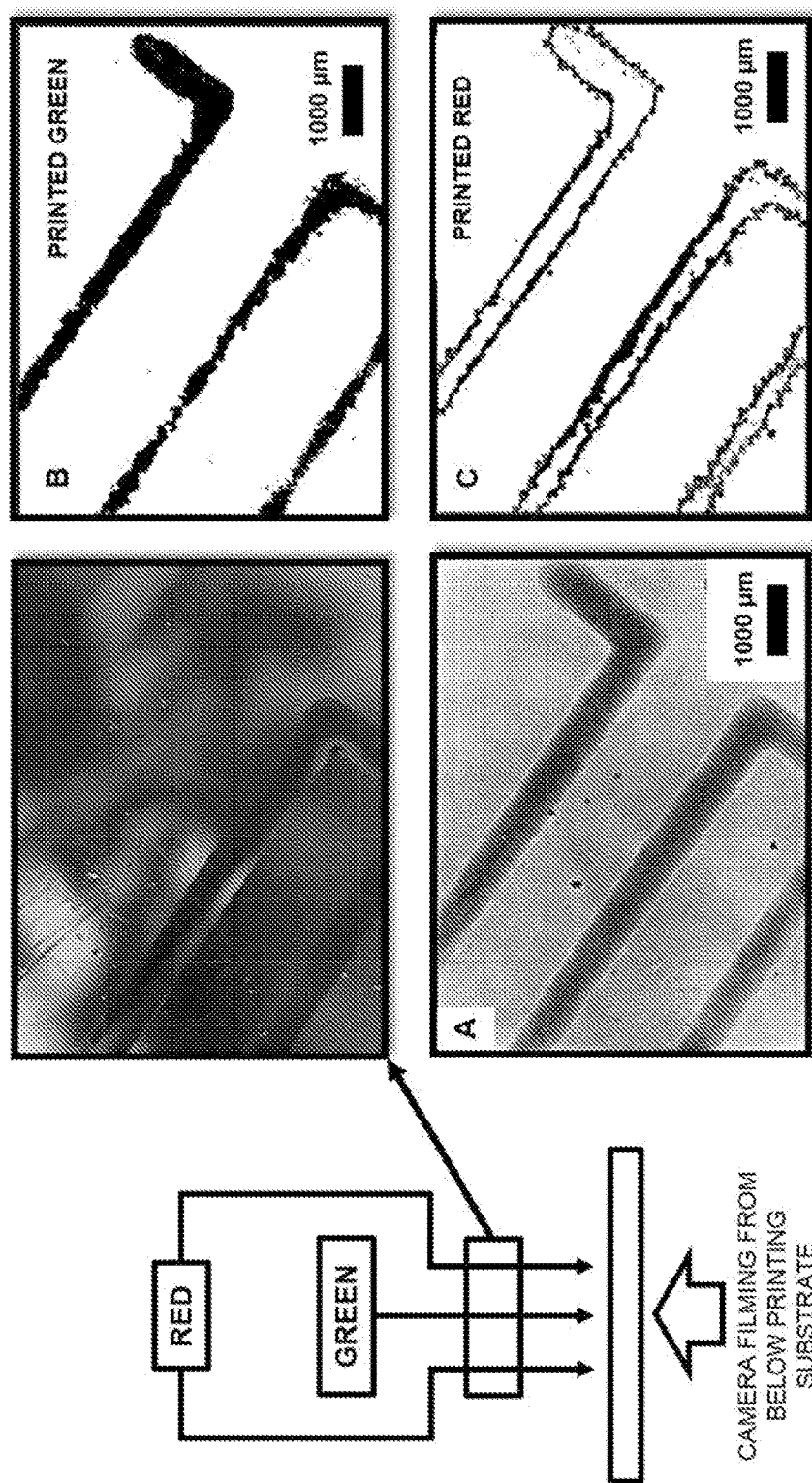


FIG. 12

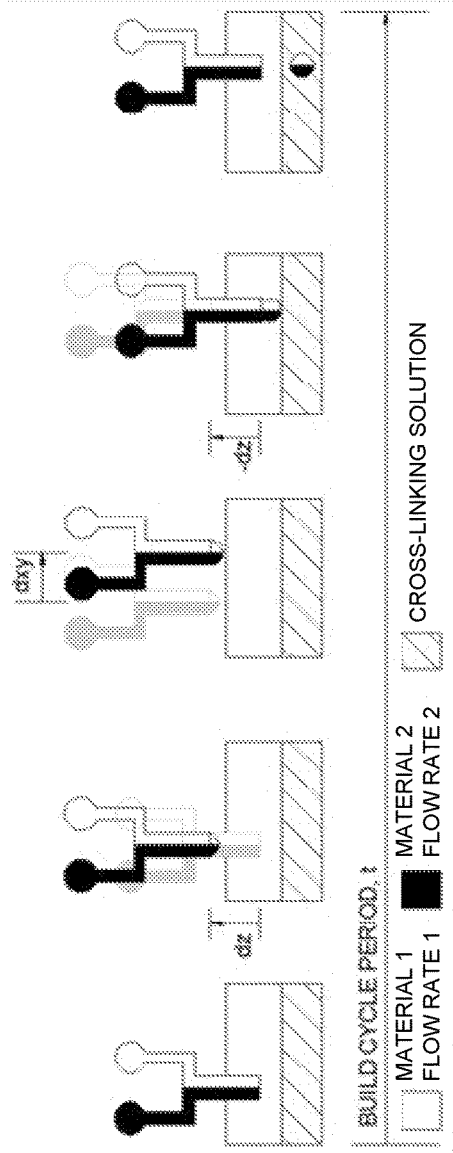


FIG. 13

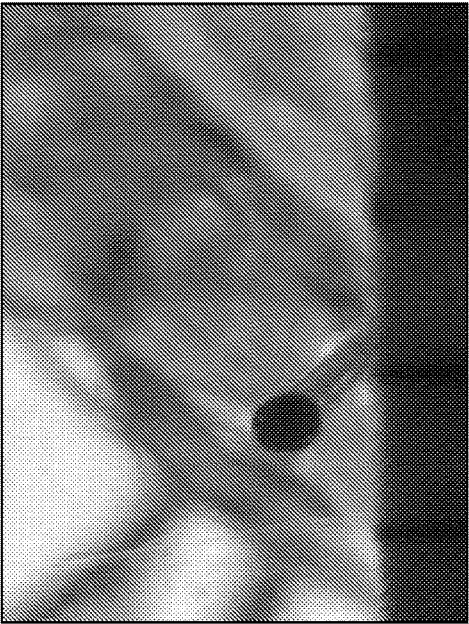


FIG. 14B

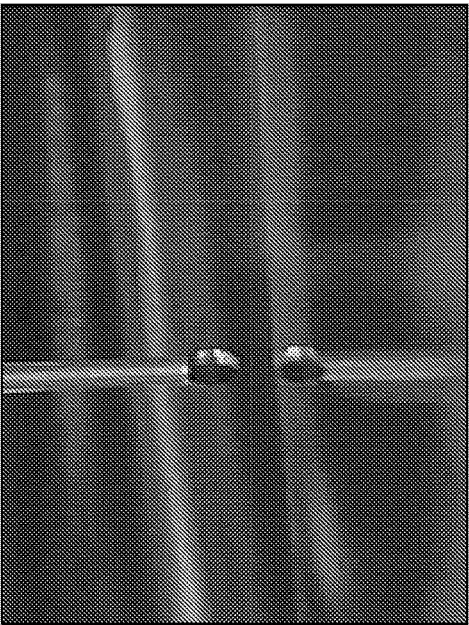


FIG. 14A

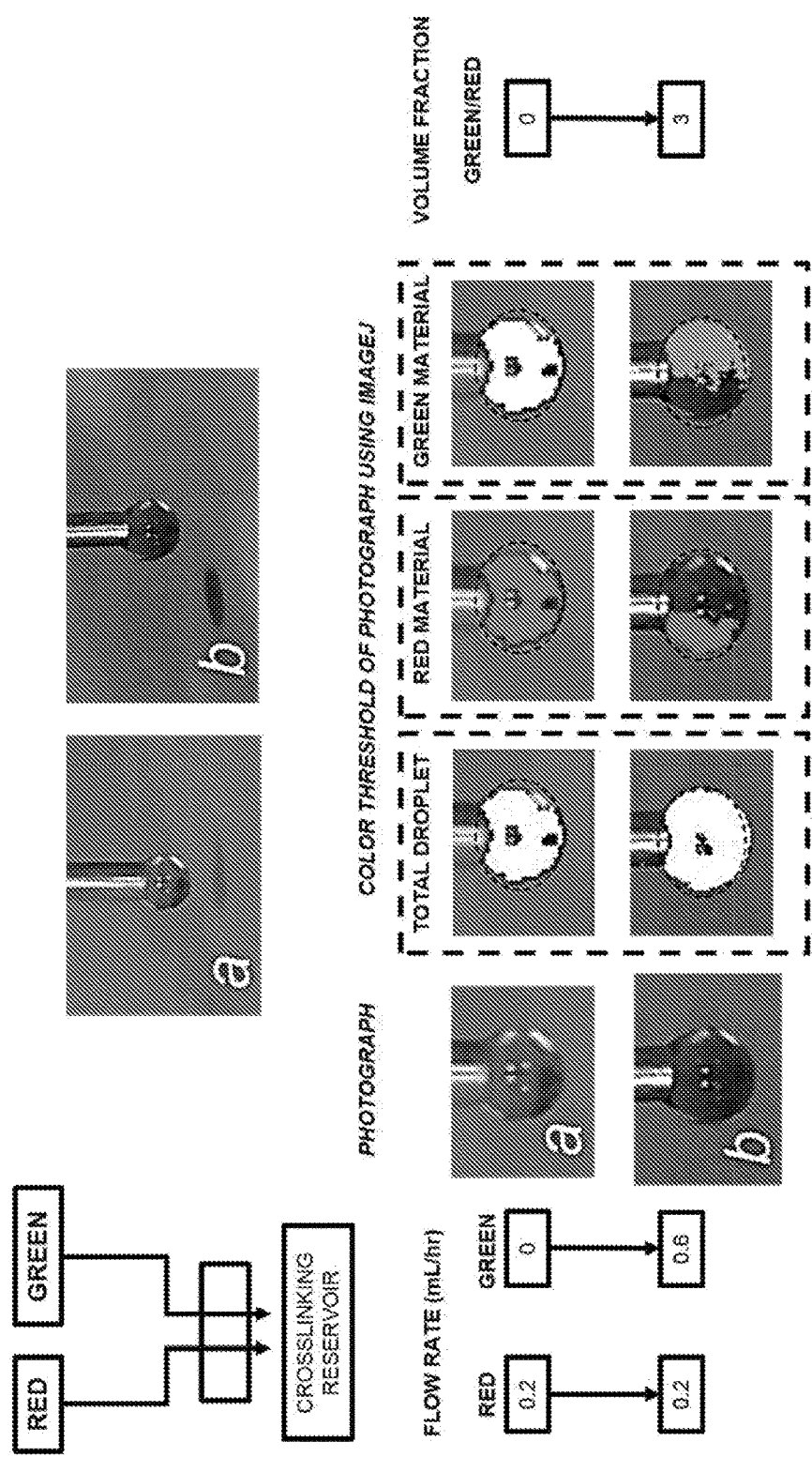


FIG. 15A

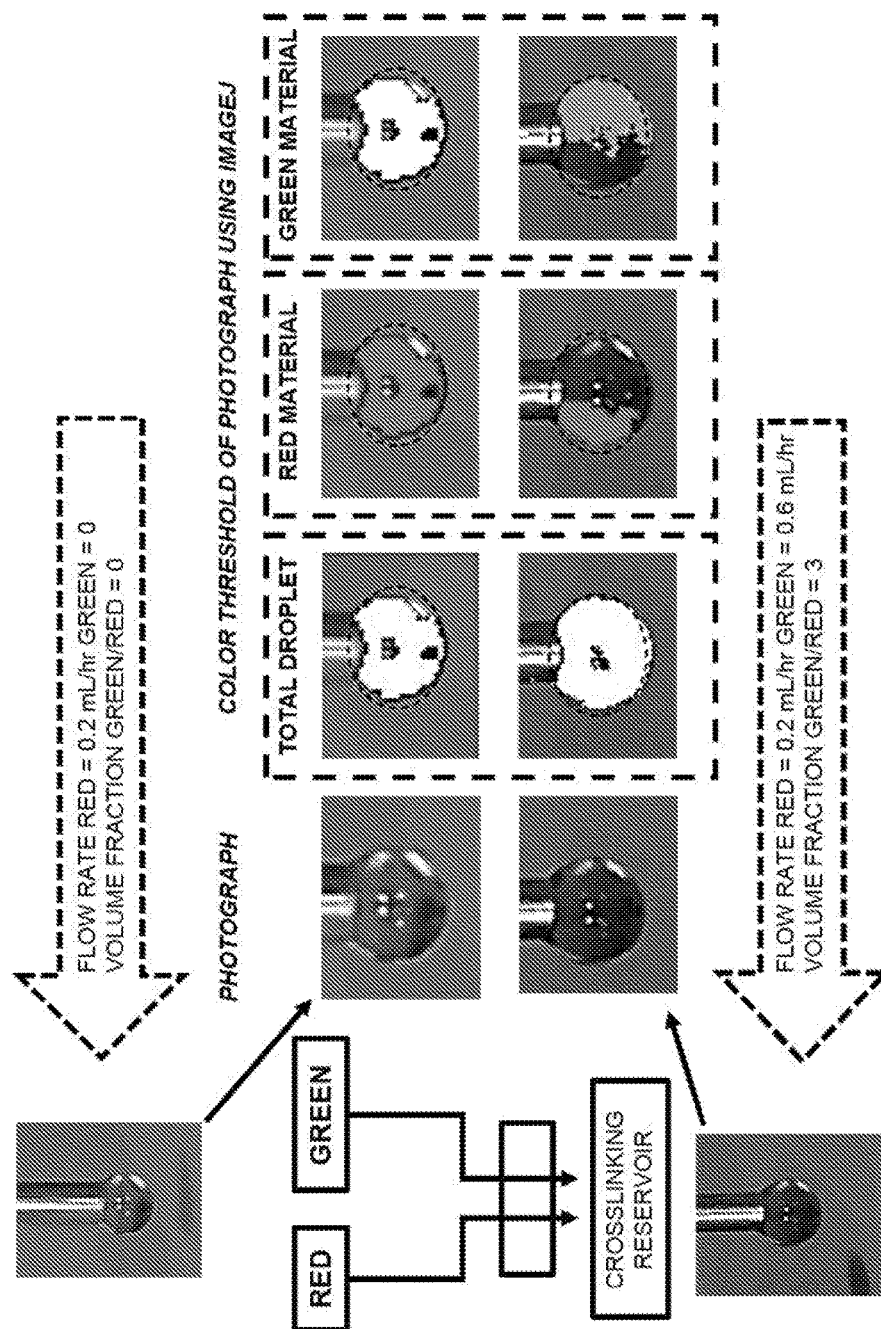


FIG. 15B

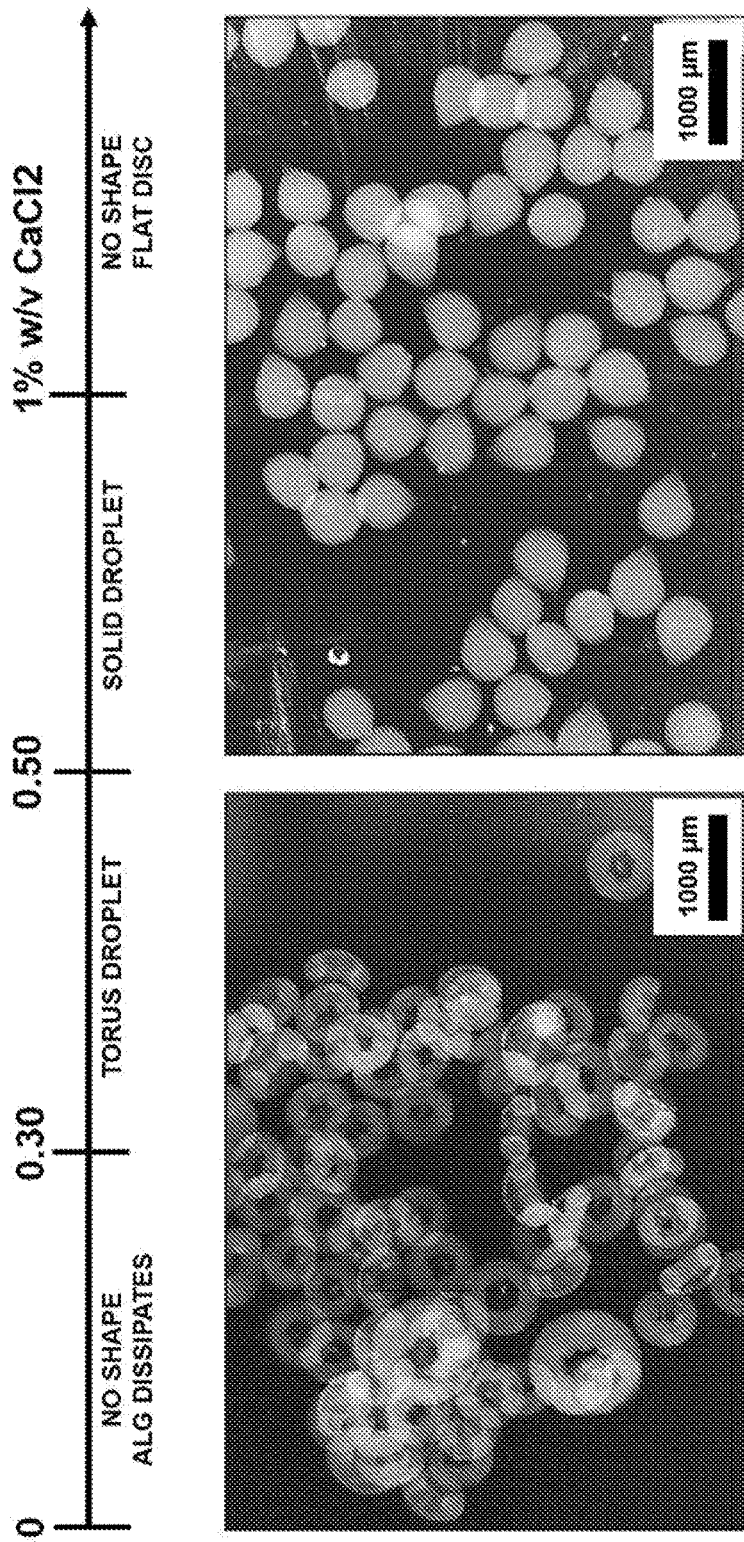


FIG. 16A

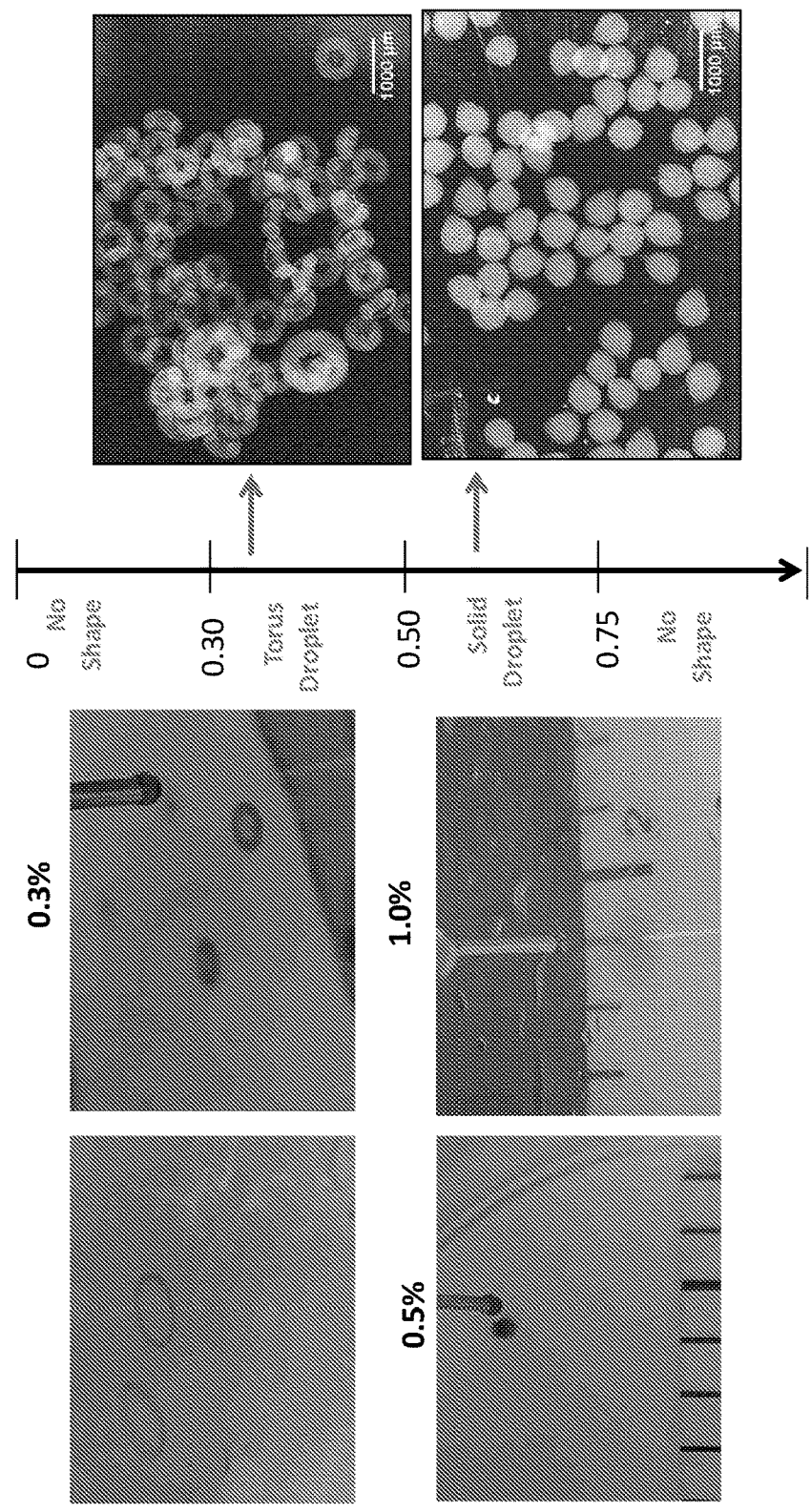


FIG. 16B

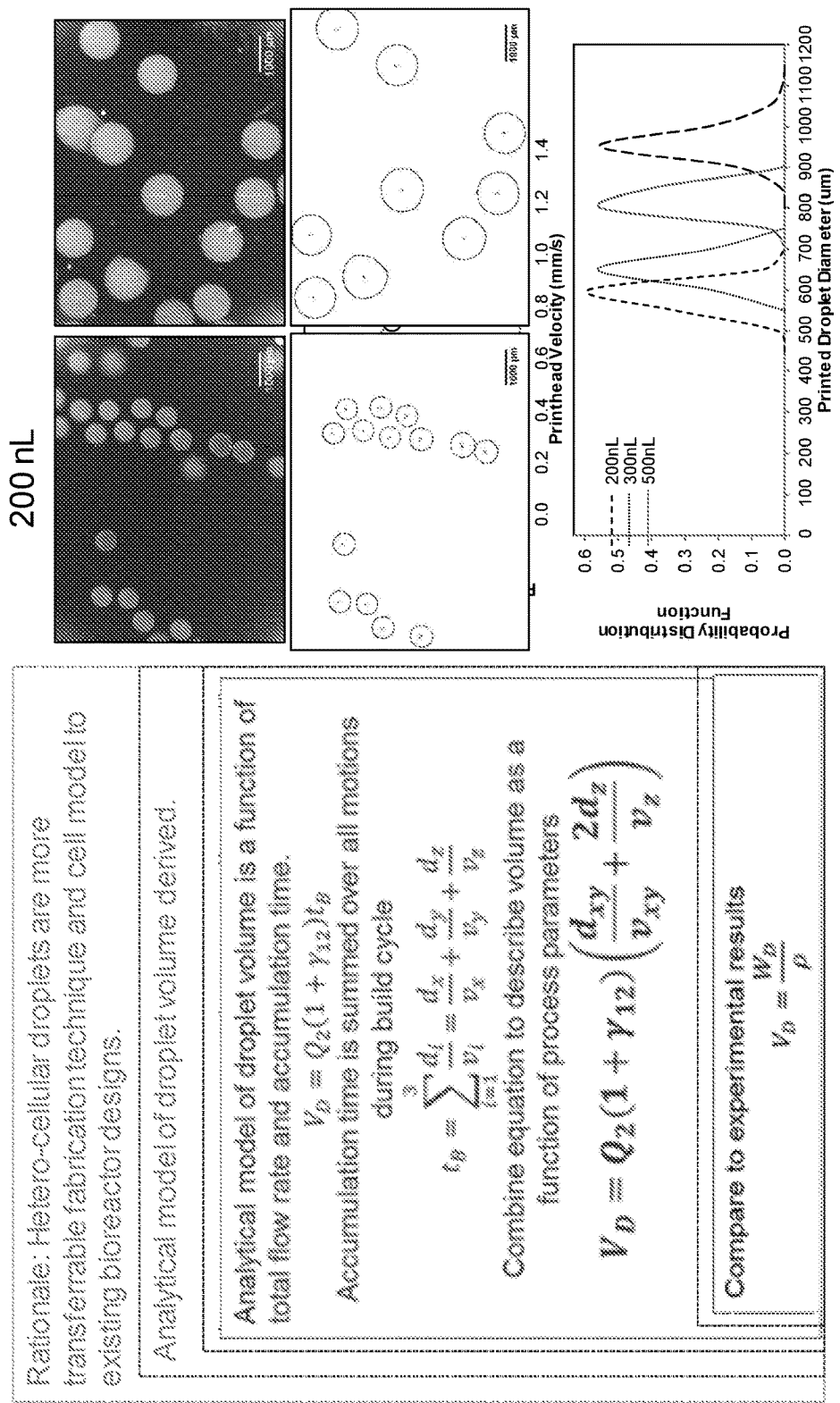


FIG. 17

**HETEROGENEOUS FILAMENTS, METHODS
OF PRODUCING THE SAME, SCAFFOLDS,
METHODS OF PRODUCING THE SAME,
DROPLETS, AND METHODS OF
PRODUCING THE SAME**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 62/197,329, filed Jul. 27, 2015. The entire content of this application is hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

[0002] The use of animal and human models is limited by the feasibility of testing protocols, availability, and ethical concerns. As a result, monolayer cell cultures are used to investigate potential anti-cancer agents. Monolayer investigations are limited because these two-dimensional (2D) models give very little feedback on the effects of the micro-environment on chemotherapeutic and the heterogeneity of the tumor.

SUMMARY OF THE INVENTION

[0003] One aspect of the invention provides a method of producing a heterogeneous filament. The method includes applying pressure to a plurality of filament components to move the filament components into a common outlet channel.

[0004] This aspect of the invention can have a variety of embodiments. The plurality of filament components can contact each other in the common outlet channel, but remain substantially unmixed. The plurality of filament components can contact each other in the common outlet channel and mix. The plurality of filament components can contact each other in the common outlet channel and mix to form one or more concentration gradients between adjacent filament components.

[0005] The plurality of filament components can be selected from the group consisting of: 2, 3, 4, 5, 6, 7, 8, 9, and 10.

[0006] The plurality of filament components can include one or more selected from the group consisting of: a polymer, a solution, a cell-laden solution, a chemically reactive solution, an aqueous solution, sodium alginate solutions, a sacrificial support material, a cell, alginate, a cross-linker, a cross-linking solution, a calcium chloride solution, and a hydrogel.

[0007] The pressure can be generated by a pump. The pump can be a syringe pump.

[0008] The heterogeneous filament can have a one-dimensional pattern. The heterogeneous filament can have a two-dimensional pattern. The heterogeneous filament can have a three-dimensional pattern.

[0009] The heterogeneous filament can be symmetrical along a longitudinal axis. The heterogeneous filament can be asymmetrical along a longitudinal axis.

[0010] The heterogeneous filament can have a largest cross-sectional dimension less than about 1 mm. The heterogeneous filament can have a largest cross-sectional dimension less than about 1 μm .

[0011] Another aspect of the invention provides a method of fabricating a scaffold. The method includes producing a plurality of filaments according to the methods described herein.

[0012] Another aspect of the invention provides a scaffold produced according to the methods described herein.

[0013] Another aspect of the invention provides a heterogeneous filament including a plurality of adjacent filament components.

[0014] This aspect of the invention can have a variety of embodiments. The plurality of filament components can contact each other in a common outlet channel, but remain substantially unmixed. The plurality of filament components can contact each other in a common outlet channel and mix. The plurality of filament components can contact each other in a common outlet channel and mix to form one or more concentration gradients between adjacent filament components.

[0015] The plurality of filament components can be selected from the group consisting of: 2, 3, 4, 5, 6, 7, 8, 9, and 10. The plurality of filament components can include one or more selected from the group consisting of: a polymer, a solution, a cell-laden solution, a chemically reactive solution, an aqueous solution, sodium alginate solutions, a sacrificial support material, a cell, alginate, a cross-linker, a cross-linking solution, a calcium chloride solution, and a hydrogel.

[0016] Another aspect of the invention provides a method of producing a droplet. The method includes: positioning a printing tip in proximity to a cross-linking liquid; applying pressure to one or more liquids to produce a droplet extending beyond outside of the printing tip; and contacting the droplet with the cross-linking liquid, thereby producing a droplet.

[0017] This aspect of the invention can have a variety of embodiments. The droplets can have a volume less than about 1 μL . The droplets can have a volume less than about 100 nL.

[0018] The cross-linking liquid can be a calcium chloride solution. The calcium chloride solution can have a concentration between about 0.3 g/L and about 0.5 g/L. The calcium chloride solution can have a concentration between about 0.5 g/L and about 1.0 g/L. The calcium chloride solution can have a concentration greater than 1.0 g/L.

[0019] The one or more liquids can include one or more selected from the group consisting of: a polymer, a solution, a cell-laden solution, a chemically reactive solution, an aqueous solution, sodium alginate solutions, a sacrificial support material, a cell, alginate, a cross-linker, a cross-linking solution, a calcium chloride solution, and a hydrogel.

[0020] The positioning step can include positioning the printing tip within about 100 μm of a surface of the cross-linking liquid.

[0021] The droplet can be a heterogeneous droplet and the applying pressure step can include applying pressure to a plurality of liquids.

[0022] The droplet can have a substantially spherical shape. The droplet can have a substantially toroidal shape. The droplet can have a substantially flat and circular shape.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] For a fuller understanding of the nature and desired objects of the present invention, reference is made to the following detailed description taken in conjunction with the

accompanying drawing figures wherein like reference characters denote corresponding parts throughout the several views.

[0024] FIG. 1 depicts microfluidic network layouts for homogenous, heterogeneous, axisymmetric, and asymmetrical combination of up to three materials according to an embodiment of the invention.

[0025] FIG. 2 depicts unit cell designs of fiber cross-sections printed according to embodiments of the invention.

[0026] FIG. 3 depicts exemplary channel networks to combine multiple materials in a 1-dimensional array according to an embodiment of the invention.

[0027] FIG. 4 depicts an exemplary block diagram of a synchronized multi-material bioprinter (SMMB) integrating a deposition head for controlled heterogeneous extrusion with a multi-nozzle deposition (MND) system according to an embodiment of the invention.

[0028] FIG. 5 is a photograph of a fabricated SMMB deposition head perfused with three different materials for visualization purposes according to an embodiment of the invention.

[0029] FIG. 6 depicts a method for fabricating an SMMB deposition head using precision extrusion deposition (PED) and replica molding processes according to an embodiment of the invention.

[0030] FIG. 7 depicts various design factors associated with printing heterogeneous cell-laden constructs using the SMMB deposition head according to an embodiment of the invention.

[0031] FIG. 8 presents photographs of a three material deposition head according to an embodiment of the invention.

[0032] FIG. 9 presents photographs of the SMMB deposition channel during volume fraction adjustments according to an embodiment of the invention.

[0033] FIG. 10 provides photographs of extrusion along a tool path to build a free-standing scaffold (left) and extrusion over a controlled build cycle to print droplets (right) according to an embodiment of the invention.

[0034] FIG. 11 provides a schematic of a synchronized multi-material deposition head mounted on a motion system printing heterogeneous line with process control of the volume fraction of each material according to an embodiment of the invention.

[0035] FIG. 12 depicts a heterogeneous printed filament in a square wave pattern printed using a bioprinter and a multi-material deposition head with two inlet channels of red and green solution according to an embodiment of the invention.

[0036] FIG. 13 is a schematic of droplet-forming process stages according to an embodiment of the invention.

[0037] FIGS. 14A and 14B are photographs of dispensing of droplets according to an embodiment of the invention.

[0038] FIGS. 15A and 15B provide photographs of red and green alginate accumulating on tip of deposition head as nano-liter droplet according to an embodiment of the invention. Photographs are color thresholded to present separate streams of red and green.

[0039] FIGS. 16A and 16B depict alginate dispersed into 0.2 g/mL calcium chloride according to an embodiment of the invention.

[0040] FIG. 17 provides an analytical model for printing heterogeneous nano-liter droplets according to an embodiment of the invention.

DEFINITIONS

[0041] The instant invention is most clearly understood with reference to the following definitions.

[0042] As used herein, the singular form “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0043] Unless specifically stated or obvious from context, as used herein, the term “about” is understood as within a range of normal tolerance in the art, for example within 2 standard deviations of the mean. “About” can be understood as within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.1%, 0.05%, or 0.01% of the stated value. Unless otherwise clear from context, all numerical values provided herein are modified by the term about.

[0044] As used in the specification and claims, the terms “comprises,” “comprising,” “containing,” “having,” and the like can have the meaning ascribed to them in U.S. patent law and can mean “includes,” “including,” and the like.

[0045] Unless specifically stated or obvious from context, the term “or,” as used herein, is understood to be inclusive.

[0046] Ranges provided herein are understood to be shorthand for all of the values within the range. For example, a range of 1 to 50 is understood to include any number, combination of numbers, or sub-range from the group consisting 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 (as well as fractions thereof unless the context clearly dictates otherwise).

DETAILED DESCRIPTION OF THE INVENTION

[0047] One aspect of the invention provides a multi-input microfluidic printhead to improve the resolution of a bioprinter by packaging multiple phenotypes in a single filament. The multi-input microfluidic printhead can be used independently or in addition to existing dispensing systems to extrude hierarchical filaments, possibly achieving greater resolution and cell seeding control.

[0048] Microfluidic techniques combine multiple material inlets in a single outlet channel without mixing due to low inertial forces in micro-scale cross-sectional channels. Embodiments of the deposition head can leverage non-mixing microfluidic flow to combine multiple materials in a heterogeneous array prior to deposition. Embodiments of the invention improve printing resolution over multi-nozzle deposition (MND) and enable mass transfer and chemical reactions, such as partial cross-linking, prior to deposition.

[0049] Embodiments of the invention can produce structures such as scaffolds through the following process. First, component materials can be selected based on composition of the target material, functional tissue, support material, and requirement for cross-linking solution. Second, the arrangement of the materials can be defined based on cell-to-cell contact in vivo and construction requirements. The volume fraction of each constituent component can be dynamically varied during the build cycle and is not restricted by the channel network in the deposition head. Third, the layout of the channel network can be defined to assemble the materials in a heterogeneous or axisymmetric style as depicted in FIG. 1.

[0050] FIG. 2 depicts unit cell designs of fiber cross-sections printed using embodiments of the invention. Fila-

ments can be composed of a variety of liquids including solutions (e.g., cell-laden, chemically reactive, and the like) and sacrificial support material to produce void spaces. The volume fraction of each solution can be tuned using the flow rate controlled by programmable syringe pumps. Assembly of solutions can be controlled by a microfluidic system. Two-dimensional microfluidic systems can produce 1-dimensional array of solutions. A more sophisticated three-dimensional microfluidic system can produce a 2-dimensional array of solutions. Exemplary channel networks to combine multiple materials in a 1-dimensional array are depicted in FIG. 3.

[0051] FIG. 4 provides an exemplary block diagram of a synchronized multi-material bioprinter (SMMB) integrating a deposition head for controlled heterogeneous extrusion with a multi-nozzle deposition (MND) system.

[0052] The design of the heterogeneous filament extruded by the SMMB can be controlled by the channel network in the deposition head. FIG. 5 is a photograph of a fabricated SMMB deposition head perfused with three different materials for visualization purposes.

[0053] FIG. 6 depicts a method for fabricating an SMMB deposition head using precision extrusion deposition (PED) and replica molding processes according to an embodiment of the invention.

[0054] FIG. 7 depicts various design factors associated with printing heterogeneous cell-laden constructs using the SMMB deposition head.

[0055] Internal features of the heterogeneous array can be constructed based on the design of the SMMB deposition head channel network and selection of the process parameter set. The design variables for the internal features can also be dynamically adjusted over the build cycle, as the process parameters can be tuned during fabrication. The width of individual or collections of adjacent material is one such controllable design variable. A parametric study swept through a set of flow rates to decrease the combined width of two materials in an axisymmetric combined outlet channel. FIG. 8 presents photographs of a three material deposition head. From left to right, the flow rate of red material is increased, while the flow rate of green and yellow is maintained constant. The combined width of green and yellow decreases from left to right. The volume fraction of green to yellow is maintained constant, as the flow rates are equal.

[0056] The volume fraction of solution can be adjusted while maintaining the combined width of the two solutions constant. FIG. 9 presents photographs of the SMMB deposition channel during volume fraction adjustments. In a second parametric study, the red flow rate is maintained constant and combined flow rate of yellow and green are maintained constant, but the volume fraction of yellow to green is adjusted. From left to right, the volume fraction of green is increased by increasing the proportion of green flow rate to yellow. The total flow rate of yellow and green is constant. The volume fraction of red is constant throughout the study, as the proportion of red to the total flow rate is constant.

[0057] Embodiments of the SMMB system described herein combine multiple materials in a heterogeneous array and deposits those materials along a pre-programmed 3-dimensional tool paths. The automation of this process offers at least two distinct opportunities for formability depicted in FIG. 10.

[0058] First, heterogamous filaments enable layer-by-layer fabrication of a free-standing scaffold or controlled cell seeding in channels of a microfluidic device deposited along a tool path. Multiple deposition heads can be simultaneously integrated in a single SMMB system. Multiple integrated deposition heads offer the same benefit as multi-nozzle printing including supplemental cross-linking solutions or secondary heterogeneous set of materials.

[0059] Second, heterogeneous nano-liter droplets can be created as described herein. Droplet volume is a controllable process parameter.

Printing Filaments

[0060] The SMMB's automated material delivery system and motion system can produce both filaments and droplets. SMMB scaffold printing improves the resolution of multi-nozzle deposition (MND) printing and offers the combined outlet channel as an opportunity for chemical reactions or mass species transfer during a controlled time interval. Partial or complete cross-linking is one method to leverage the combined outlet channel. SMMB droplet printing produces a simpler product than scaffold printing. However, printed droplets are a more accessible and adaptable technology. Droplets are simpler to design, fabricate, and ensure sufficient nutrient diffusion to the cell-laden core. Droplets can also be used in many existing bioreactor platforms. In one embodiment, printed co-culture nano-liter droplets are the model of human liver to study the effect of microgravity on cell morphology and drug up-take using a rotary cell culture system.

[0061] The synchronized multi-material bioprinter (SMMB) dispensing system can be integrated with the existing multi-nozzle deposition (MND) system to continuously extrude heterogeneous filaments along a tool path. SMMB system performance to pattern a heterogeneous filament along a tool path can be defined by a set of geometric design variables to control printed filament cross-sectional width/height, reaction time in the outlet channel. Design variables $\Psi_{SMMB,D}$ can be defined by droplet volume, V_D , and volume fraction of material i to material j , γ_{ij} .

[0062] FIG. 11 provides a schematic of a synchronized multi-material deposition head mounted on a motion system printing heterogonous line with process control of the volume fraction of each material. FIG. 11 also depicts the internal architecture of an exemplary printed filament.

[0063] FIG. 12 depicts a heterogeneous printed filament in a square wave pattern printed using a bioprinter and a multi-material deposition head with two inlet channels of red and green solution. A schematic of the microfluidic network in multi-material deposition head depicts the red and green flows combined in a single outlet channel. The bioprinter's motion system carries the deposition head over the stationary substrate to produce the square wave pattern. A photograph of the printed alginate filament in square wave pattern (A) and binary contrast enhancement of green (B) and red (C) alginate are shown.

Printing Droplets

[0064] Solutions can be prepared and loaded into the SMMB material delivery system syringe pump for controlled extrusion during the build cycle. In this work, sodium alginate aqueous solutions of alginic acid sodium salt from brown algae and a cross-linking solution of ACS grade

calcium chloride were prepared from distilled water, respectively. Alginate solution was loaded into the SMMB material delivery system and a cross-linking solution flooded the reservoir pool positioned below the printing tip. The SMMB deposition head outlet tip was lowered to less than 100 μm above cross-linking pool. The programmable syringe pump extruded alginate through the deposition head. The alginate accumulates on the tip, until contact with the cross-linking reservoir. On contact, the droplet is released from the tip and gels to form an alginate pool in the cross-linking reservoir. The build cycle period, or time to printing one droplet, can be adjusted using the process parameters to fabricate variable size droplets. FIG. 13 is a schematic of a the process stages. FIGS. 14A and 14B are photographs of dispensing of droplets.

[0065] Referring now to FIGS. 15A and 15B, material accumulates on the tip of the deposition head as a nano-liter droplet. Photographs of the accumulating droplet can be analyzed for mixing. Homogenous droplets present red coloring. The heterogeneous droplets present both red and green coloring. The two colors are presented on discrete sides of the droplet.

[0066] The success of droplet formability requires that the alginate gel in the cross-linking bath before dispersing in the bath due to the inertial force of the printing process. The inertial forces of the flowing fluid degrades the droplet's spherical structure before gelation if the cross-linking concentration is low. Aqueous alginate solution will flow until sufficient time in contact with cross-linking solution causes gelation. Further, printed droplet solidity (e.g., torus or sphere) are generated by controlled the cross-linking time. Cross-linking time is a function of the alginate and calcium chloride concentration. Cross-linking concentration can be controlled between about 0.2 g/mL and 1.0 g/mL calcium chloride in distilled water. Alginate concentration can be constant 0.5% (w/v) throughout the study. The length of time required for cross-linking can be inversely related to the concentration of the crossing solution, for a given concentration of alginate. Time lapse photography of the droplet dispensing into the cross-linking pool presents the effect of cross-linking solution concentration on formability of droplets and the ability to produce torus structures. FIGS. 16A and 16B depicts alginate dispersed into 0.2 g/mL calcium chloride. The printed material flows away from the deposition point and degrades any printed structure. Further perturbation of the cross-linking reservoir will cause the droplet to twist and lose all recognizable shape.

[0067] FIG. 17 provides an analytical model for printing heterogeneous nano-liter droplets.

Remarks

[0068] The novel synchronized multi-material bioprinter integrates microfluidic techniques with 3D cell printing to package multiple cell-laden materials and cross-linking solutions along a tool path or as a nanoliter droplet. Biomimetic assembly to support cell viability in vitro and solicit paracrine/autocrine signaling between cells presents methodological progress to guide the cell aggregate to perform tissue-level function; with application to tissue engineering and general built biological constructs. Bioprinting is an enabling technology to engineer built biological systems. Interfacing biology and architecture in built biological systems with reproducibility and engineering process control requires advanced manufacturing. Bioprinting is a computer

aided manufacturing method with process control over (1) macro-scale (10^{-3} - 10^{-1} m) architecture and (2) micro-scale (10^{-4} - 10^{-5} m) heterogeneous packaging of components and internal features. Macro-scale architecture is critical if the built biological system is designed to physically fit into a larger system. Micro-scale heterogeneous packaging of organisms and support artifacts is critical to the function of the built biological system. Additional artifacts heterogeneously packed with the biology during printing spatial-temporally effect the built system's mechanics, physics, and chemistry. Artifact candidates include load bearing structural elements, open porous networks for diffusion, functionalized or magnetic nanoparticles, and piezoelectric/conductive components.

Implementation in Computer-Readable Media and/or Hardware

[0069] The methods described herein can be readily implemented in software that can be stored in computer-readable media for execution by a computer processor. For example, the computer-readable media can be volatile memory (e.g., random access memory and the like) non-volatile memory (e.g., read-only memory, hard disks, floppy disks, magnetic tape, optical discs, paper tape, punch cards, and the like).

[0070] Additionally or alternatively, the methods described herein can be implemented in computer hardware such as an application-specific integrated circuit (ASIC).

EQUIVALENTS

[0071] Although preferred embodiments of the invention have been described using specific terms, such description is for illustrative purposes only, and it is to be understood that changes and variations may be made without departing from the spirit or scope of the following claims.

INCORPORATION BY REFERENCE

[0072] The entire contents of all patents, published patent applications, and other references cited herein are hereby expressly incorporated herein in their entireties by reference.

[0073] Additionally, Applicant notes and hereby incorporates by reference herein the following publications by Applicant describing 3D printing devices: U.S. Pat. No. 8,639,484 and U.S. Patent Application Publication Nos. 2006/195179, 2008/020049, 2008/145639, 2009/263849, 2011/136162, 2011/165646, 2011/177590, and 2012/080814.

1. A method of producing a heterogeneous filament, the method comprising:

applying pressure to a plurality of filament components to move the filament components into a common outlet channel.

2. The method of claim 1, wherein the plurality of filament components contact each other in the common outlet channel, but remain substantially unmixed.

3. The method of claim 1, wherein the plurality of filament components contact each other in the common outlet channel and mix.

4. The method of claim 1, wherein the plurality of filament components contact each other in the common outlet channel and mix to form one or more concentration gradients between adjacent filament components.

5. The method of claim 1, wherein the plurality of filament components is selected from the group consisting of: 2, 3, 4, 5, 6, 7, 8, 9, and 10.

6. The method of claim 1, wherein the plurality of filament components include one or more selected from the group consisting of: a polymer, a solution, a cell-laden solution, a chemically reactive solution, an aqueous solution, sodium alginate solutions, a sacrificial support material, a cell, alginate, a cross-linker, a cross-linking solution, a calcium chloride solution, and a hydrogel.

7. The method of claim 1, wherein the pressure is generated by a pump.

8. The method of claim 7, wherein the pump is a syringe pump.

9. The method of claim 1, wherein the heterogeneous filament has a one-dimensional pattern.

10. The method of claim 1, wherein the heterogeneous filament has a two-dimensional pattern.

11. The method of claim 1, wherein the heterogeneous filament has a three-dimensional pattern.

12. The method of claim 1, wherein the heterogeneous filament is symmetrical along a longitudinal axis.

13. The method of claim 1, wherein the heterogeneous filament is asymmetrical along a longitudinal axis.

14. (canceled)

15. (canceled)

16. A method of fabricating a scaffold, the method comprising:

producing a plurality of filaments according to the method of claim 1.

17. A scaffold produced according to the method of claim 1.

18. A heterogeneous filament comprising:

a plurality of adjacent filament components.

19. The heterogeneous filament of claim 18, wherein the plurality of filament components contact each other in a common outlet channel, but remain substantially unmixed.

20. The heterogeneous filament of claim 18, wherein the plurality of filament components contact each other in a common outlet channel and mix.

21. The heterogeneous filament of claim 18, wherein the plurality of filament components contact each other in a common outlet channel and mix to form one or more concentration gradients between adjacent filament components.

22. (canceled)

23. (canceled)

24. A method of producing a droplet, the method comprising:

positioning a printing tip in proximity to a cross-linking liquid;

applying pressure to one or more liquids to produce a droplet extending beyond outside of the printing tip;

and

contacting the droplet with the cross-linking liquid;

thereby producing a droplet.

25-36. (canceled)

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